Oceanic Tracers

Learning objectives:

- Understand the difference between classes/types of oceanic tracers
- Introduce the history of global ocean surveys
- Learn to perform water-mass mixing calculations
- Describe and use quasi-conservative geochemical tracers

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Type	Use
Natural - Stable	
Natural - Radioactive	
Anthropogenic - Transient	
Anthropogenic - Deliberate	

Type	Use
Natural - Stable	
O_2	Water mass analysis, non-conservative
PO_4, NO_3, SiO_2	Nutrients high in Pacific water, low in Atlantic Water
NO, PO, PO_4^*	Water mass analysis, quasi-conservative
^{18}O, ^4He, Ne	Identify and quantify glacial ice (^4He & Ne high, ^{18}O low in glacial ice)
^3He	Circulation of CDW onto Antarctic continental shelves (^3He high in CDW); gas exchange
Natural - Radioactive	
^{14}C	Large scale deep water circulation and mixing; ocean CO_2 uptake on century to millennial time scales
Anthropogenic - Transient	
CFC-11, CFC-12, CFC-113, CCl_4, SF_6	Identify most recently formed water; circulation pathways, water mass ages, water mass formation rates, estimation of anthropogenic CO_2 uptake
^3H	Same as trace gases above, but weak signal in S.H.
Bomb ^{14}C	Ocean uptake of anthropogenic CO_2
Anthropogenic - Deliberate	
SF_6, CF_3SF_5	Mixing, circulation, and gas exchange

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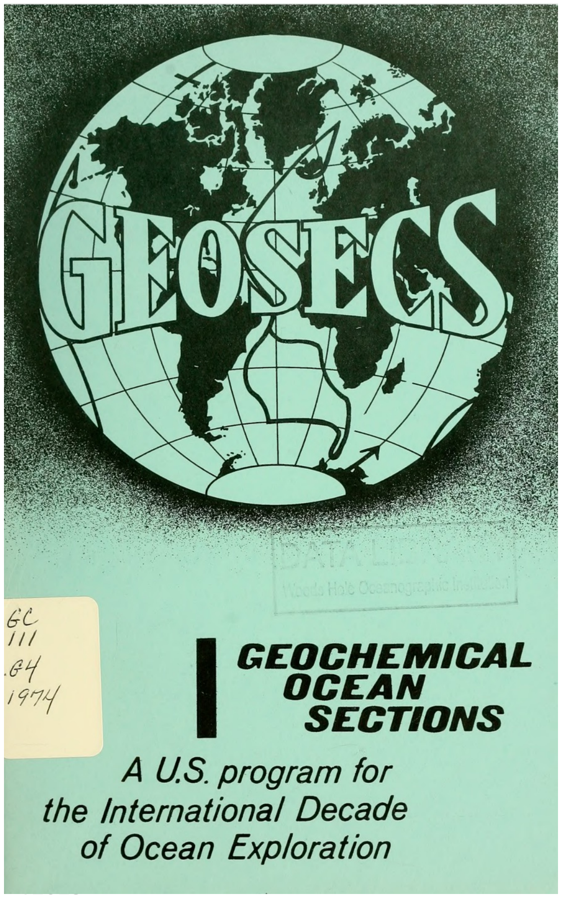
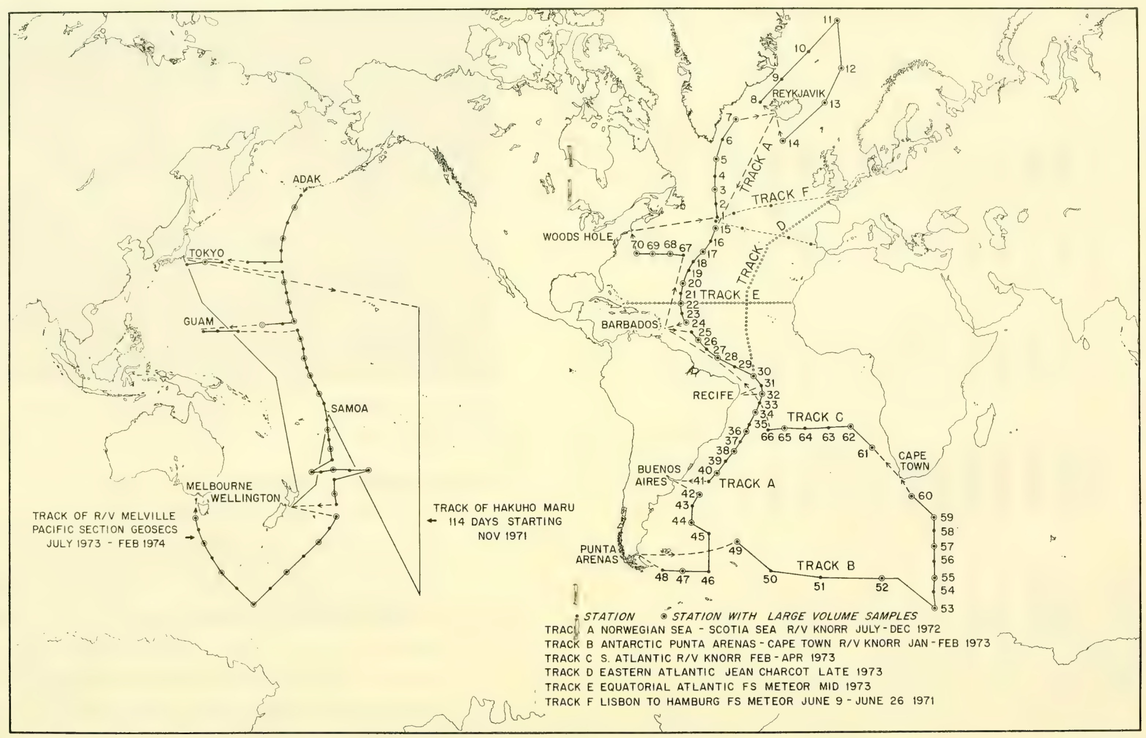
GEochemical Ocean SECtions Study (GEOSECS)





OCT 1, 1971

PROPOSED GEOSECS CRUISE TRACKS

Figure 4

Atlantic from Jul 1972 to May 1973
Pacific from Aug 1973 to Jun 1974
Indian from Dec 1977 to Mar 1978

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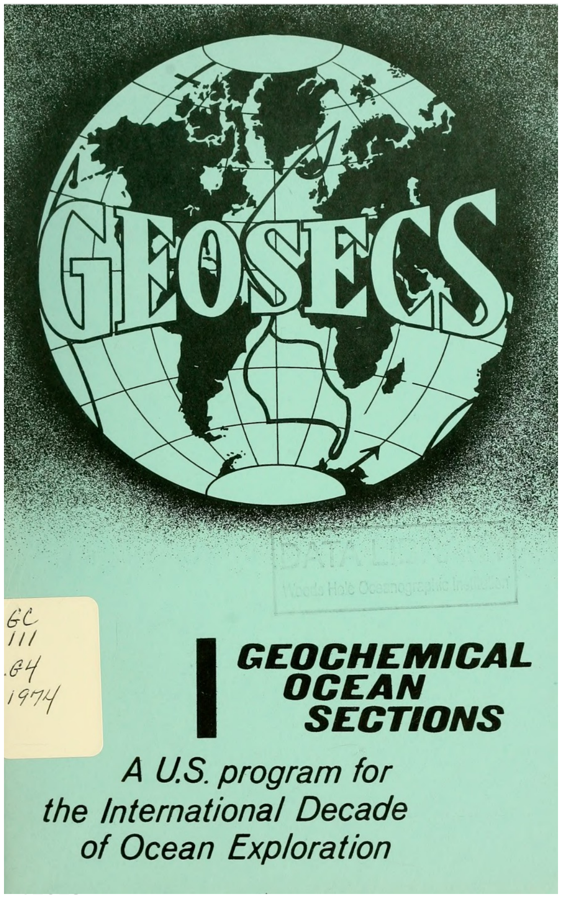
GEochemical Ocean SECtions Study (GEOSECS)



WHAT IS GEOSECS?

As man becomes increasingly aware of the ocean as a source of food, a disposal area for nuclear and industrial waste products, a strategic realm for national security and a controlling factor in the earth's climatic regime, he also recognizes how very little he knows about the sea. Until recently, oceanography has been a science of exploration, of mapping variables that are relatively easy to determine such as temperature and total salt concentration and of studying the pattern of wind-driven currents in the surface layers of the water. Yet, the ultimate use of the sea depends critically upon a detailed understanding of other processes in the deep sea -- the interchange of material between deep and surface water, the variation of organic productivity throughout the oceans and the exchange of water and gases with the atmosphere.

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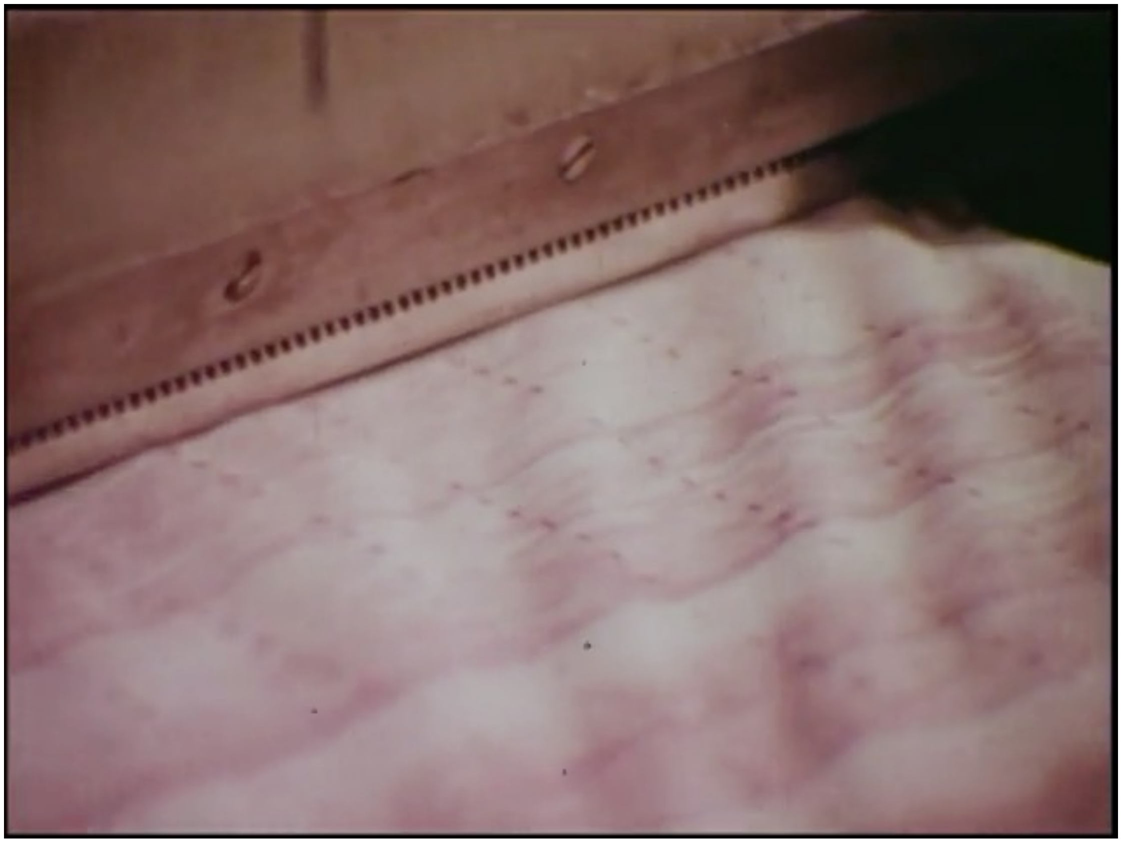
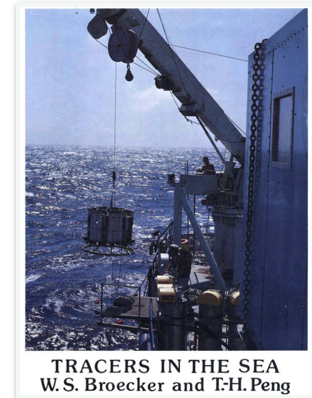
GEOSECS



From NSF educational film, "Rivers in the Sea"



"Although GEOSECS made no astounding discoveries, our superb global data set became the basis for a host of subsequent research projects. I benefited greatly; it became the grist for my tome "Tracers in the Sea" which, 30 years later, remains the book on the geochemistry of the oceans."

-Wally Broecker, doi:10.7185/geochempersp

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https://www.ldeo.columbia.edu/~broecker/Home_files/TracersInTheSea_searchable.pdf

Transient Tracers in the Oceans (TTO)

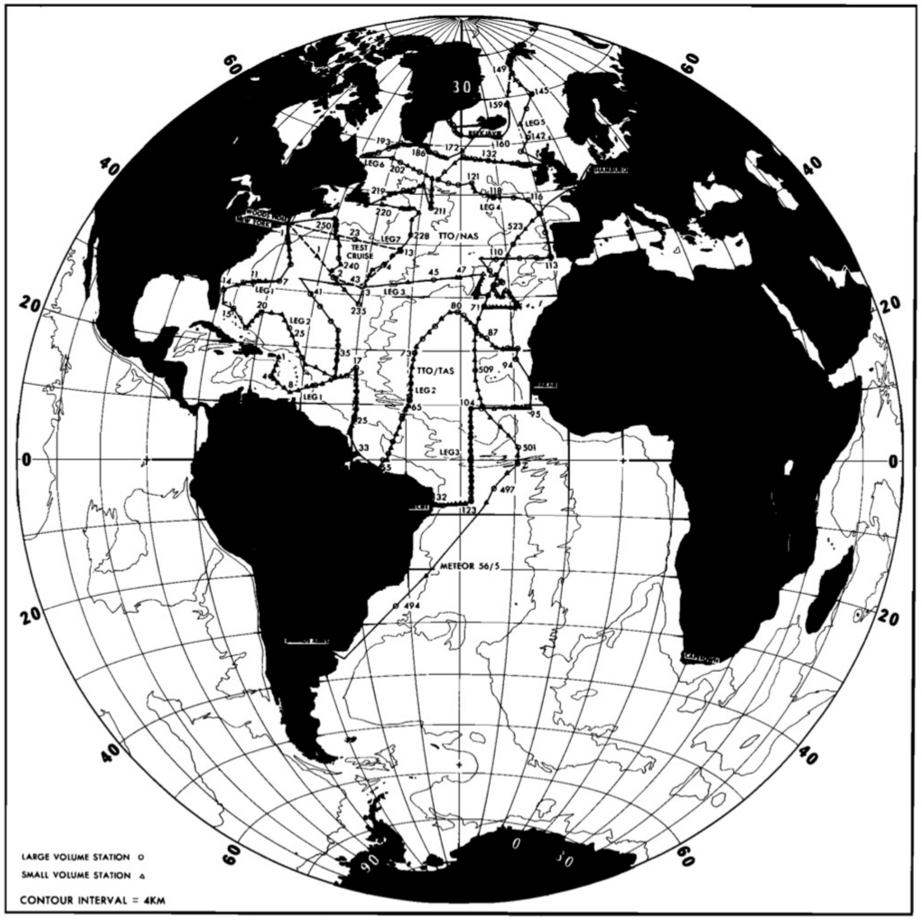


Fig. 1. Cruise track and station locations for the Transient Tracers in the Ocean program.

Brewer et al., 1985

JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 90, NO. C4, PAGES 6903–6905, JULY 20, 1985

The Transient Tracers in the Ocean (TTO) Program: The North Atlantic Study, 1981; The Tropical Atlantic Study, 1983

PETER G. BREWER

Woods Hole Oceanographic Institution, Massachusetts

JORGE L. SARMIENTO

Geophysical Fluid Dynamics Program, Princeton University, New Jersey

WILLIAM M. SMETHIE, JR.

Lamont-Doherty Geological Observatory of Columbia University, Palisades, New York

The expeditions, sponsored by the National Science Foundation and the U.S. Department of Energy (North Atlantic only), were designed to observe the passage of man-made geochemical tracers into the interior of the ocean. The foundations for such an experiment were laid in the 1972–1978 GEOSECS program. Here, for the first time, a systematic survey revealed the penetration into the thermocline and deep ocean of the products of man's military/industrial activities, principally tritium and carbon-14 resulting from atmospheric testing of nuclear weapons, which terminated with the nuclear test ban treaty in 1962.

U.S. JGOFS – Joint Global Ocean Flux Study

U.S. JGOFS: History and Mission

The U.S. JGOFS program, a component of the U.S Global Change Research Program, grew out of the recommendations of a National Academy of Sciences workshop in 1984. The international program, which has more than 30 participating nations, began three years later under the auspices of the Scientific Committee on Oceanic Research (SCOR). In 1989, JGOFS became a core program of the International Geosphere-Biosphere Programme (IGBP). JGOFS has two primary goals:

1. To determine and understand on a global scale the processes controlling the time-varying fluxes of carbon and associated biogenic elements in the ocean and to evaluate the related exchanges with the atmosphere, sea floor and continental boundaries;
2. To develop a capability to predict on a global scale the response of oceanic biogeochemical processes to anthropogenic perturbations, in particular those related to climate change.

The strategy for addressing these goals has five major components:

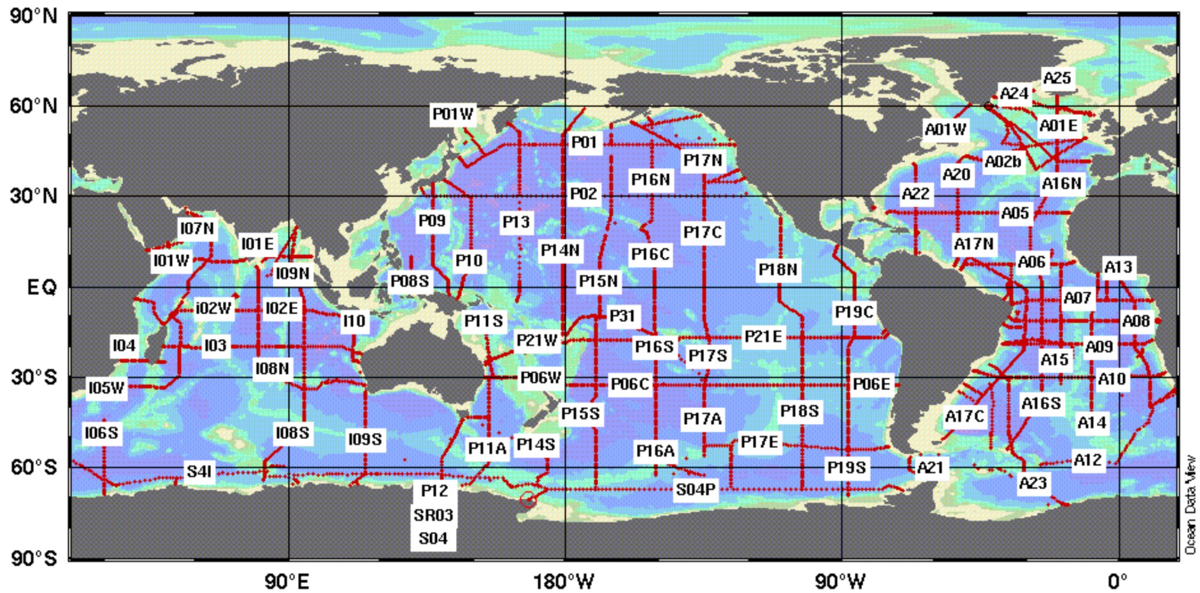
1. a global survey of oceanic CO2 and the bio-optical properties of the surface ocean, coordinated with the World Ocean Circulation Experiment (WOCE);
2. long-term time-series observations at key oceanic sites;
3. a series of studies of biogeochemical processes in parts of the ocean that lend themselves to effective observation of particular phenomena;
4. the development of models to assimilate results, produce large-scale descriptions and predict responses to future disturbances;
5. the development of an accessible, comprehensive biogeochemical database.

World Ocean Circulation Experiment (WOCE)

OCADS > Access Data > WOCE Program Data

WOCE Program Data

Click on section for data and information



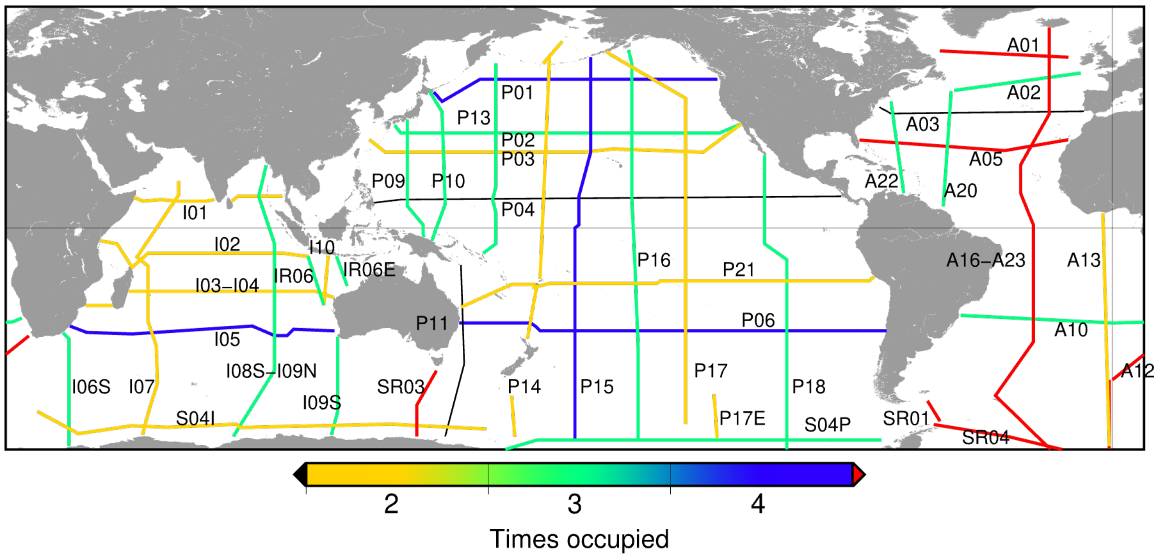
https://www.ncei.noaa.gov/access/ocean-carbon-acidification-data-system/oceans/CDIACmap.html https://woceatlas.ucsd.edu/

http://woce.nodc.noaa.gov/wdiu/updates/Data_Resource.pdf

WOCE AND ITS OBSERVATIONS

The Hydrographic Programme was one part of the global sampling effort within WOCE, which also included satellite observations of the ocean surface, measurements of ocean currents using surface drifters, subsurface floats, current meter moorings, acoustic Doppler current profilers, measurements of sea level using tide gauges, repeated surveys for temperature using expendable bathythermographs, and surface meteorology measurements (see Siedler, Church and Gould 2001). WOCE also supported major modelling projects, including general circulation models of both the ocean alone and of the ocean coupled with the atmosphere, and ocean data assimilation activities. It had links to many other programmes such as the Joint Global Ocean Flux Study (JGOFS) (Wallace, 2001) and the Tropical Ocean and Global Atmosphere (TOGA) Observing System (Godfrey et al., 2001). The WOCE field programme took approximately ten years to complete, but most observations were carried out between 1990 and 1998. The synthesis and modelling components of WOCE and the wider scientific exploitation of WOCE results will continue for many years.

GO-SHIP

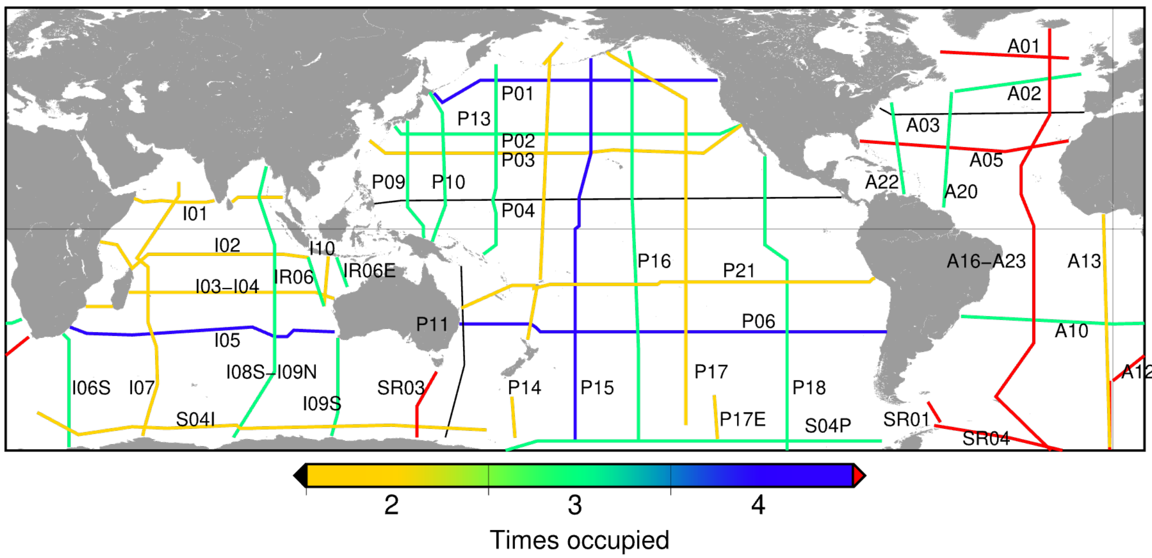


https://cchdo.ucsd.edu/products/goship-easyocean

"GO-SHIP brings together scientists with interests in physical oceanography, the carbon cycle, marine biogeochemistry and ecosystems, and other users and collectors of hydrographic data to develop a globally coordinated network of sustained hydrographic sections as part of the global ocean/climate observing system."

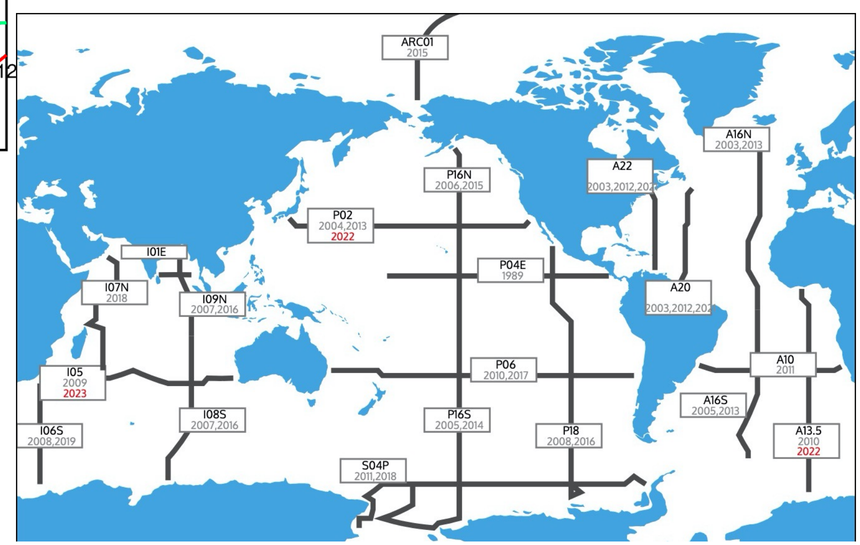
- International GO-SHIP mission statement

GO-SHIP



https://cchdo.ucsd.edu/products/goship-easyocean

Cruises for the U.S. Global Ocean Carbon and Repeat Hydrography Program, 2003 - 2023
(Red indicates a pending cruise, grey indicates a completed cruise)



https://usgoship.ucsd.edu/usgoship.ucsd.edu/hydromap/

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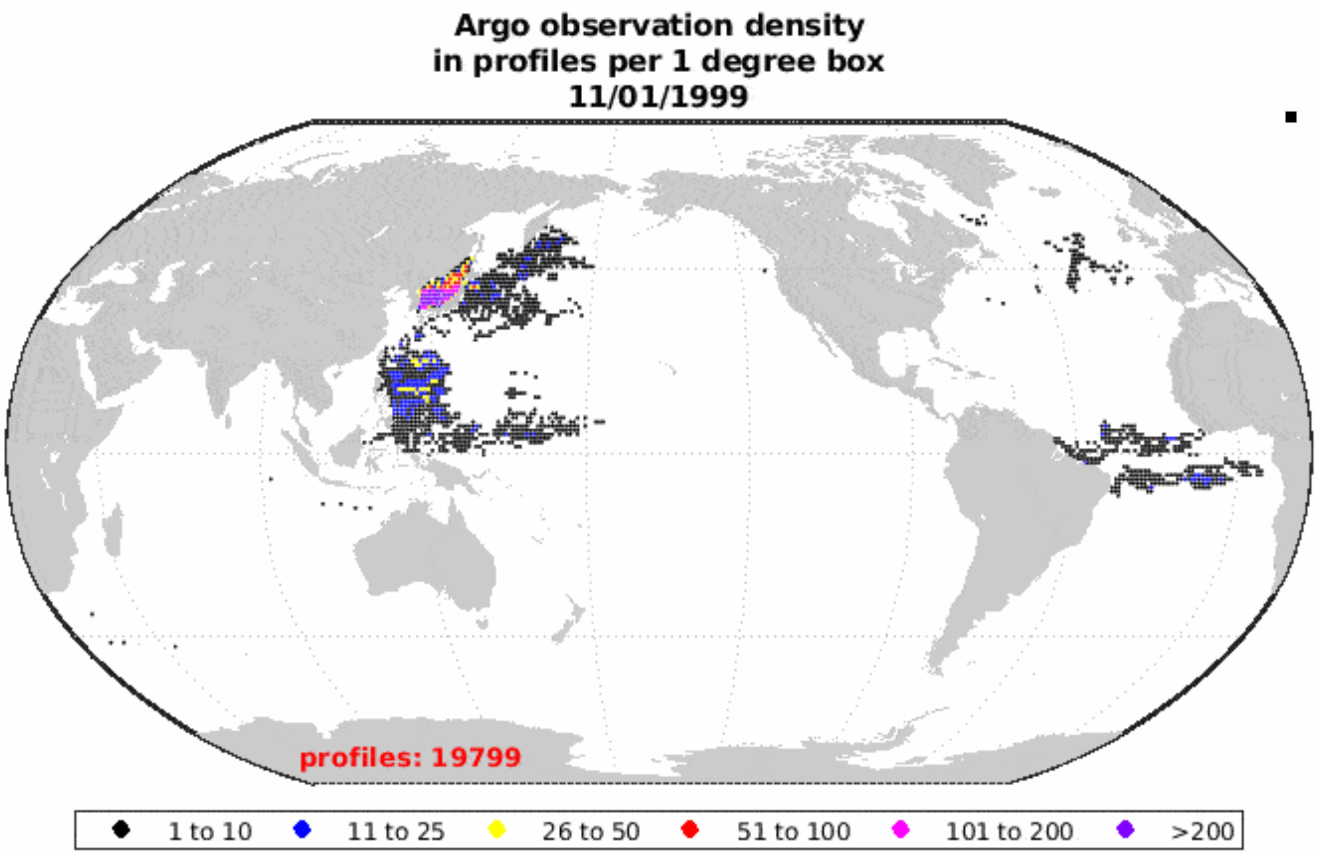
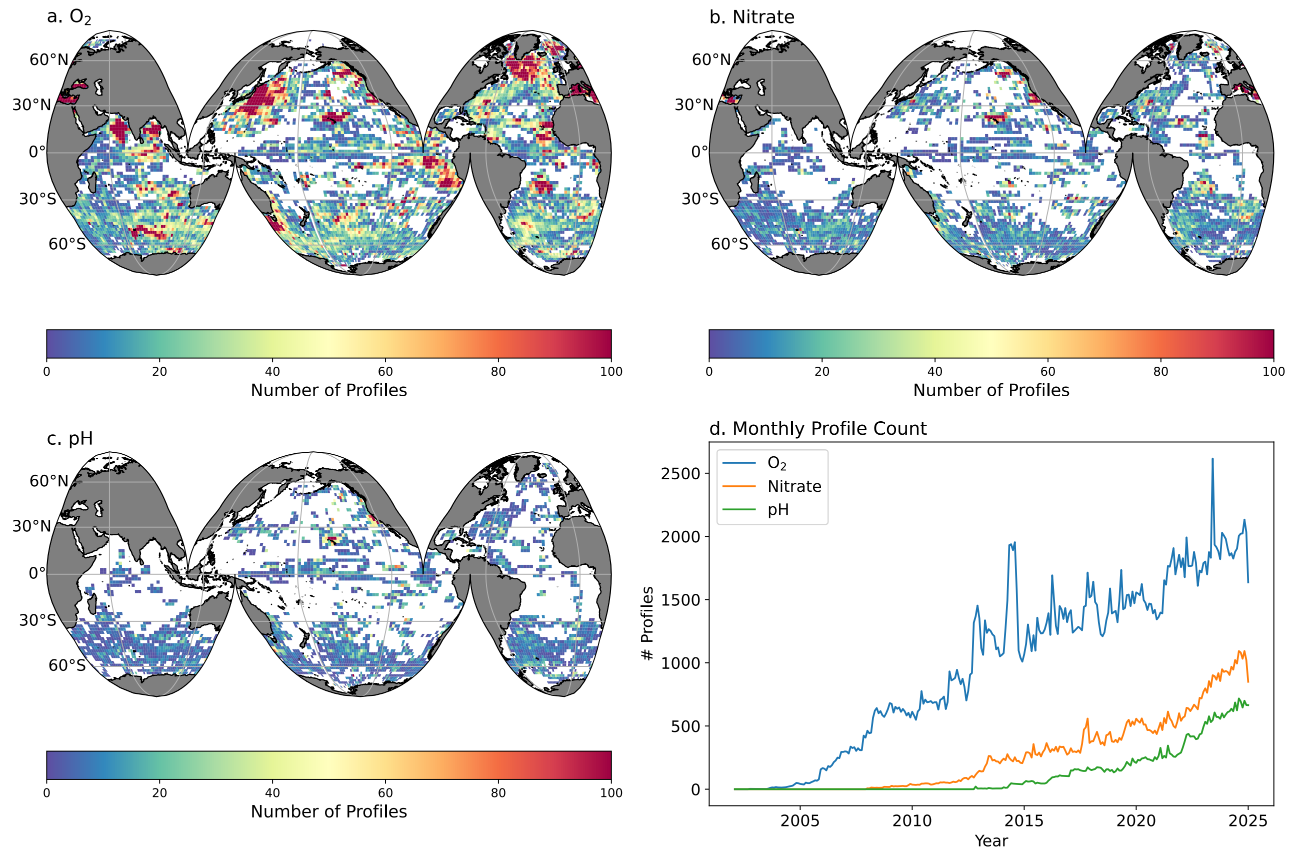
Argo – global profiling floats



- Temperature, Salinity, Pressure

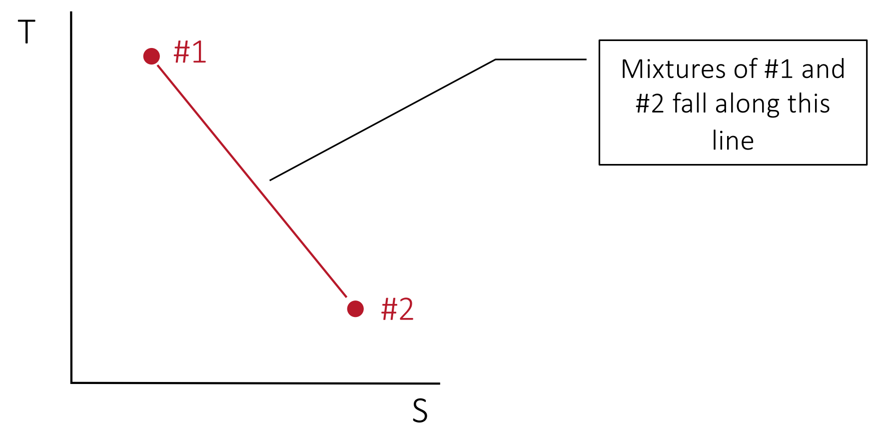
Biogeochemical Argo



- Temperature, Salinity, Pressure
- Oxygen, nitrate, pH
- Bio-optical measurements

Mixing of Two Water Masses

- Plot two stable **conservative** tracers for each water mass on an X-Y plot (e.g., T, S)
- Link points with a linear "mixing line"
- Water samples with compositions falling along the mixing line are composed of mixtures of the two water masses



Water mass mixing definitions

- Assume: conservation of mass, heat, salt
- Definitions:
  - C_i = concentration of some stable conservative property in water mass "i" (end-member concentration)
  - C_{mix} = concentration of some stable **conservative** property in a water mixture
  - f_i = fraction of the water mixture that is from water mass "i"
  - f_{mix} = $1.0 = f_1 + f_2 + \ldots + f_i$ (mass balance equation)

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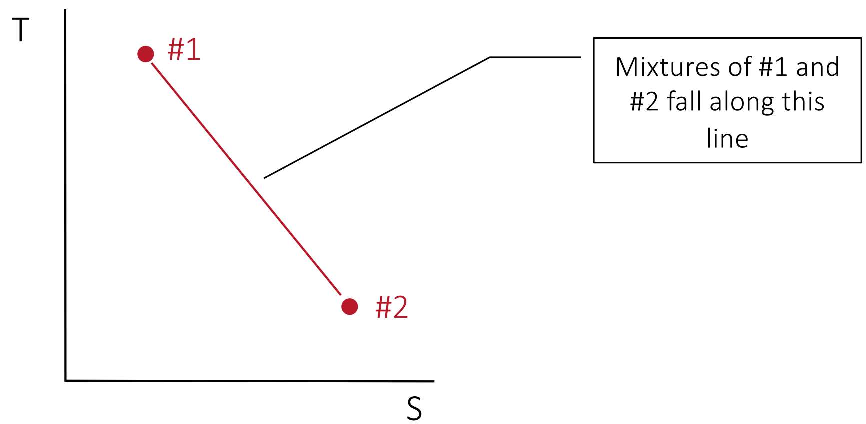

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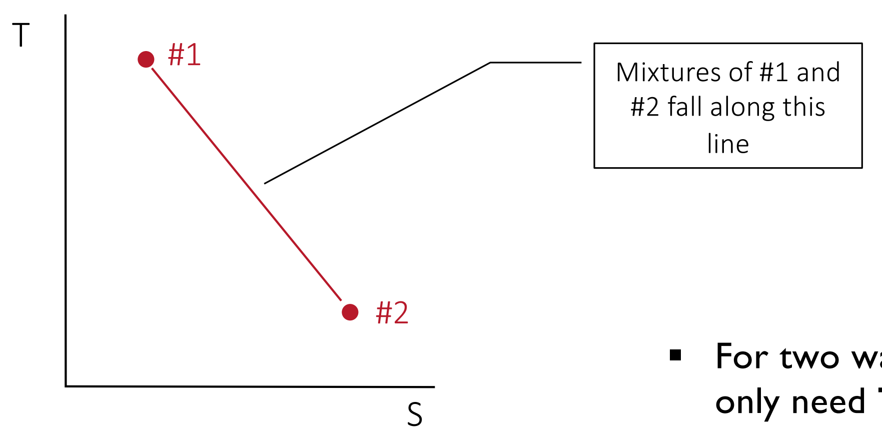

$$1 = f_1 + f_2 \quad (1)$$

$$C_{mix} = f_1 C_1 + f_2 C_2 \quad (2)$$

Combining:

$$f_1 = \frac{C_{mix} - C_2}{C_1 - C_2}$$

- For two water masses: only need **T** or **S**

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Mixing of Four Water Masses

We need four equations with four variables. For example:

$$1 = f_1 + f_2 + f_3 + f_4$$

$$T_{mix} = f_1T_1 + f_2T_2 + f_3T_3 + f_4T_4$$

$$S_{mix} = f_1S_1 + f_2S_2 + f_3S_3 + f_4S_4$$

$$?_{mix} = f_1?_1 + f_2?_2 + f_3?_3 + f_4?_4$$

Thus, we need three stable, conservative tracers:

T_i = temperature of water mass "i"
T_{mix} = temperature of a water mixture

S_i = salinity of water mass "i"
S_{mix} = salinity of a water mixture

$?_i$ = [?] of water mass "i"
$?_{mix}$ = [?] of a water mixture

- What to use for the third conservative tracer?

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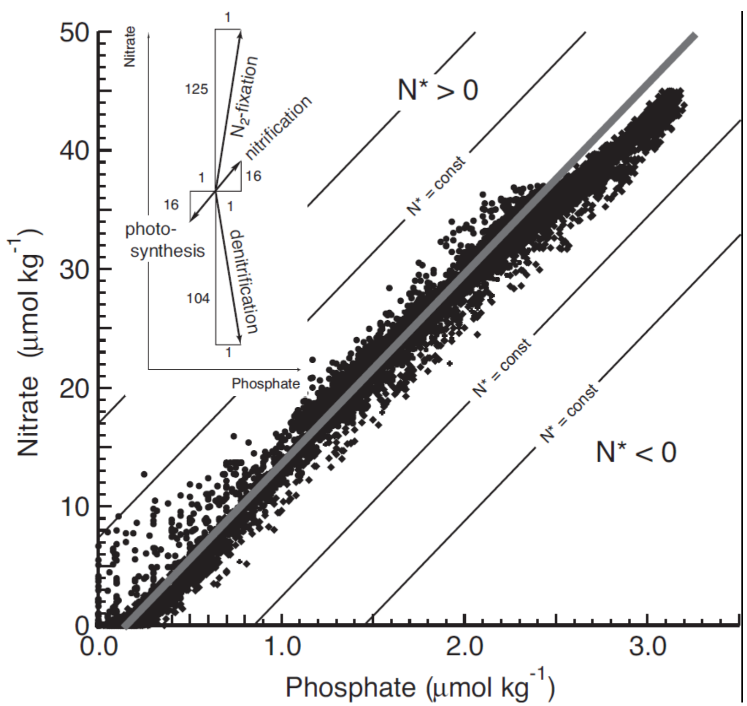
Quasi-conservative tracers (from Export lecture)

N* is a tracer for denitrification in the ocean



N* is defined as
$N^* = [NO_3] - 16 \times [PO_4] + 2.9$

The solid line shows the linear equation with N:P =16

Values to the right have negative N* (denitrification) to the left have positive N* (nitrogen fixation)

PO_4 versus Nitrate (WOCE data) from Sarmiento and Gruber (2006)

"NO" – A Quasi-Conservative Water Mass Tracer

$106\ CO_2 + 16\ HNO_3 + H_3PO_4 + 122\ H_2O +$ trace elements (e.g. *Fe, Zn, Mn…*) $\rightarrow$

$$(CH_2O)_{106}(NH_3)_{16}(H_3PO_4) + 138\ O_2$$

"NO" – A Quasi-Conservative Water Mass Tracer

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$$(CH_2O)_{106}(NH_3)_{16}(H_3PO_4) + 138\ O_2$$

$\Delta O_2 / \Delta N = -138 / 16 = -8.6$

Thus, during the oxidation of organic matter:

For each mole of NO_3^- released to a water mass,

~9 moles of O_2 is removed

Broecker (1974): "NO" $\equiv 9[NO_3^-] + [O_2]$

"PO" $\equiv 135[PO_4^-] + [O_2]$

Earth and Planetary Science Letters, 23 (1974) 100–107
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"NO", A CONSERVATIVE WATER-MASS TRACER*

WALLACE S. BROECKER
Lamont-Doherty Geological Observatory and Department of Geology, Columbia University, Palisades, N.Y. (USA)

Received February 20, 1974
Revised version received June 13, 1974

A composite property, $9NO_3 + O_2$, is proposed as a conservative water-mass tracer. The coefficient 9 is chosen so that the increase in "NO" resulting from nitrate introduction during respiration just balances the consumption of dissolved oxygen gas. Because of the pronounced difference in the preformed nitrate content of deep water produced at the northern and southern ends of the Atlantic Ocean "NO" provides an independent means of disentangling the degree of mixing of various water types in the deep sea. Evidence based on data obtained during the Atlantic Geosecs program is presented to demonstrate the sensitivity and reliability of this conservative tracer.

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Conservative Behavior of "NO"

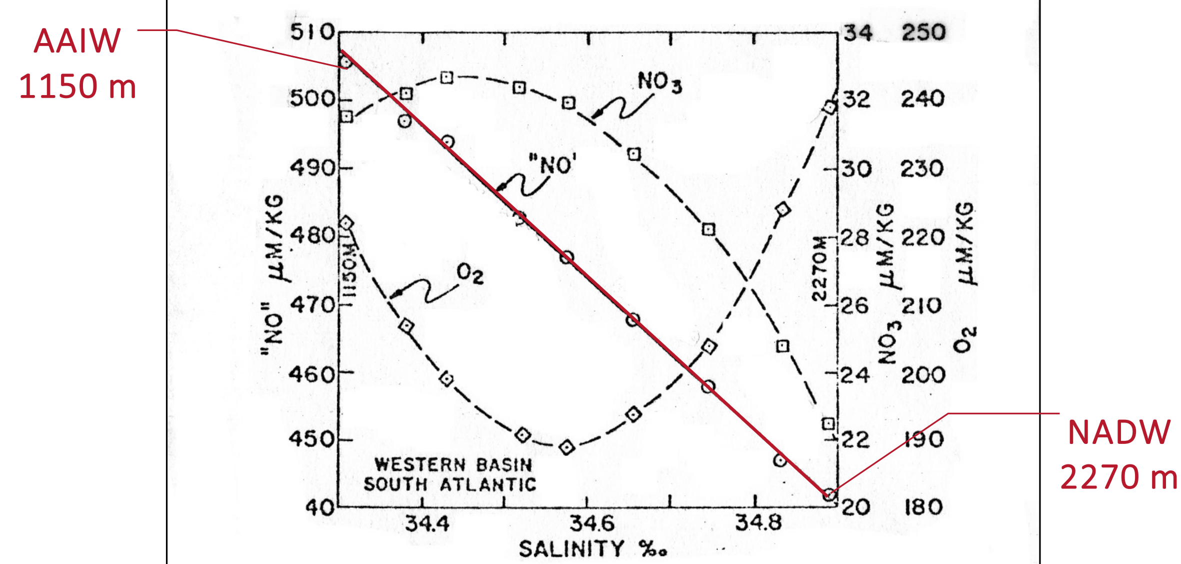

Fig. 5. "NO", NO_3, and O_2 versus salinity at Atlantic Geosecs station 60 in the western basin of the South Atlantic between the core of the AAIW (1150 m) and the top of the NADW (2270 m). The excess NO_3 content is just balanced by the O_2 deficiency yielding a straight line relationship between salinity and "NO".

Broecker, 1974

Mixing of Four Water Masses

We need four equations with four variables. For example:

$$1 = f_1 + f_2 + f_3 + f_4$$

$$T_{mix} = f_1T_1 + f_2T_2 + f_3T_3 + f_4T_4$$

$$S_{mix} = f_1S_1 + f_2S_2 + f_3S_3 + f_4S_4$$

$$NO_{mix} = f_1NO_1 + f_2NO_2 + f_3NO_3 + f_4NO_4$$

Thus, we need three stable, conservative tracers:

T_i = temperature of water mass "i"
T_{mix} = temperature of a water mixture

S_i = salinity of water mass "i"
S_{mix} = salinity of a water mixture

NO_i = NO of water mass "i"
NO_{mix} = NO of a water mixture

PO_4* – Another Quasi-Conservative Water Mass Tracer

GLOBAL BIOGEOCHEMICAL CYCLES, VOL. 5, NO. 1, PAGES 87-117, MARCH 1991

RADIOCARBON DECAY AND OXYGEN UTILIZATION IN THE DEEP ATLANTIC OCEAN

Wallace S. Broecker, Sean Blanton, and William M. Smethie, Jr.

Lamont-Doherty Geological Observatory of Columbia University, Palisades, New York

Gote Ostlund

Rosenstiel School of Marine and Atmospheric Sciences, University of Miami, Miami, Florida

Abstract. A parameter based on the sum of the concentrations of PO_4 and O_2 (divided by the Redfield coefficient $-\Delta O_2/\Delta PO_4$) is used to separate the contributions of the northern and southern components to deep waters in the Atlantic. This separation allows the amount of radiocarbon lost by radiodecay and the amount of oxygen lost to respiration during residence in the deep Atlantic to be calculated. Maps of these quantities reveal strong west to east gradients and weak north to south gradients consistent with ventilation along the western boundary from both ends of the ocean coupled with mixing outward from the boundary. The O_2 and ^{14}C deficiences are highly correlated, suggesting an O_2 utilization rate of 12 µm/kg per century. The apparent mean isolation time of water in the deep Atlantic is about 200 years.

INTRODUCTION

Both the distribution of dissolved oxygen and that of radiocarbon in the deep Atlantic are strongly influenced by the mixing between the northern and southern components of deep water. Hence if the extent of changes in O_2 content caused by respiration and of changes in the radiocarbon to carbon ratio caused by radioactive decay are to be isolated, then a means must be found to accurately define the relative contribution of these two sources to any given water sample. As was shown by Broecker [1979] and by Broecker et al. [1985], a promising means to do this is through the use of a PO_4-O_2 based quasi-conservative parameter. Broecker [1974] initially suggested a parameter called "PO" based on the sum of the O_2 concentration and a Redfield ratio multiple (i.e., 132) of the PO_4 concentration. In this paper we employ a modification of "PO" which we believe yields a more accurate separation of the contributors. We apply this new separation procedure to the expansion of the Geochemical Ocean Sections (GEOSECS) data set for the deep Atlantic made available by the Transient Tracers in the Ocean (TTO), Tropical Atlantic Survey (TAS) and South Atlantic Ventilation Experiment (SAVE).

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Paper number 90GB02279.
0886-6236/91/90GB-02779$10.00

COMPONENT CHARACTERIZATION

We base our separation of the northern and southern component contributions to deep waters (i.e., >2000 m) in the Atlantic on a parameter we designate as PO_4*. It is related to the measured PO_4 and O_2 concentrations as follows:

$$PO_4^* = PO_4 + \frac{O_2}{175} - 1.95 \mu m/kg$$

The constant term (1.95) is introduced in order to bring the PO_4* values into the range of measured PO_4 contents in the deep sea. The revised Redfield ratio (175) relating O_2 consumption to PO_4 production during respiration is the global average proposed by Broecker et al. [1985]. It is based on the analysis of water column chemical data from six different regions in the ocean deemed suitable for this type of analysis. As is summarized in Table 1, despite the

Broecker (1985):

$\Delta O_2 / \Delta PO_4 = -175 / 1 = -175$

Thus, during the oxidation of organic matter:

For each mole of PO_4^- released to a water mass, ~175 moles of O_2 is removed

Broecker et al. (1991):

$$\boxed{[PO_4^*] \equiv [PO_4^-] + \frac{[O_2]}{175} - 1.95\ \mu mol/kg}$$

Separating NADW from AABW

Salinity (S) is conservative, but 4 water masses that combine to form NADW have different S.

PO_4^* is nearly uniform within AABW and within the four components of NADW (*Broecker et al.*, 1998):

AABW: 1.95 ± 0.07 μmol kg^{-1}

NADW: 0.73 ± 0.07 μmol kg^{-1}

Fraction of NADW is:

$$f_{NADW} = \frac{1.95 - [PO_4^*]}{1.95 - 0.73}$$

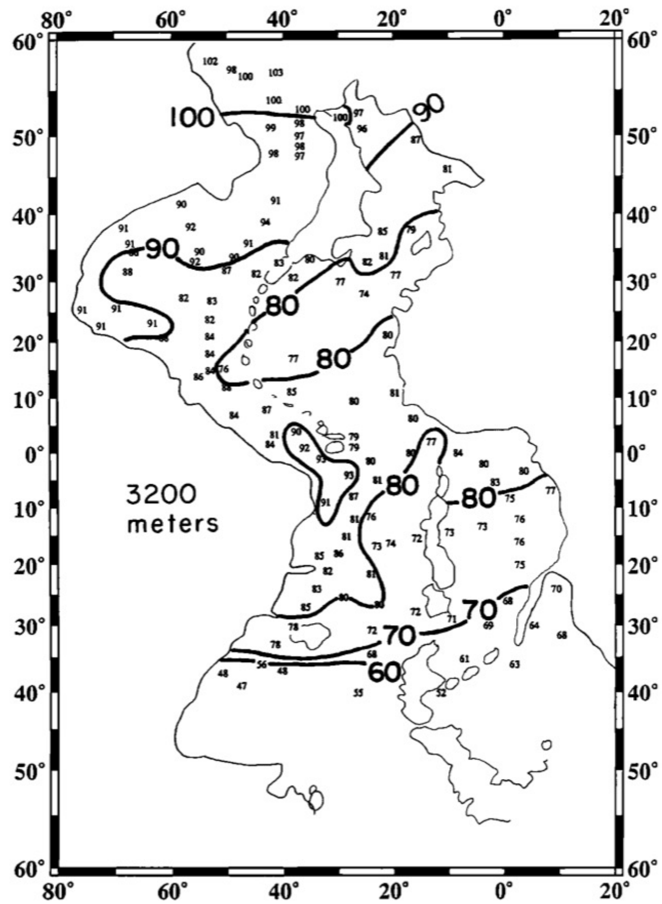
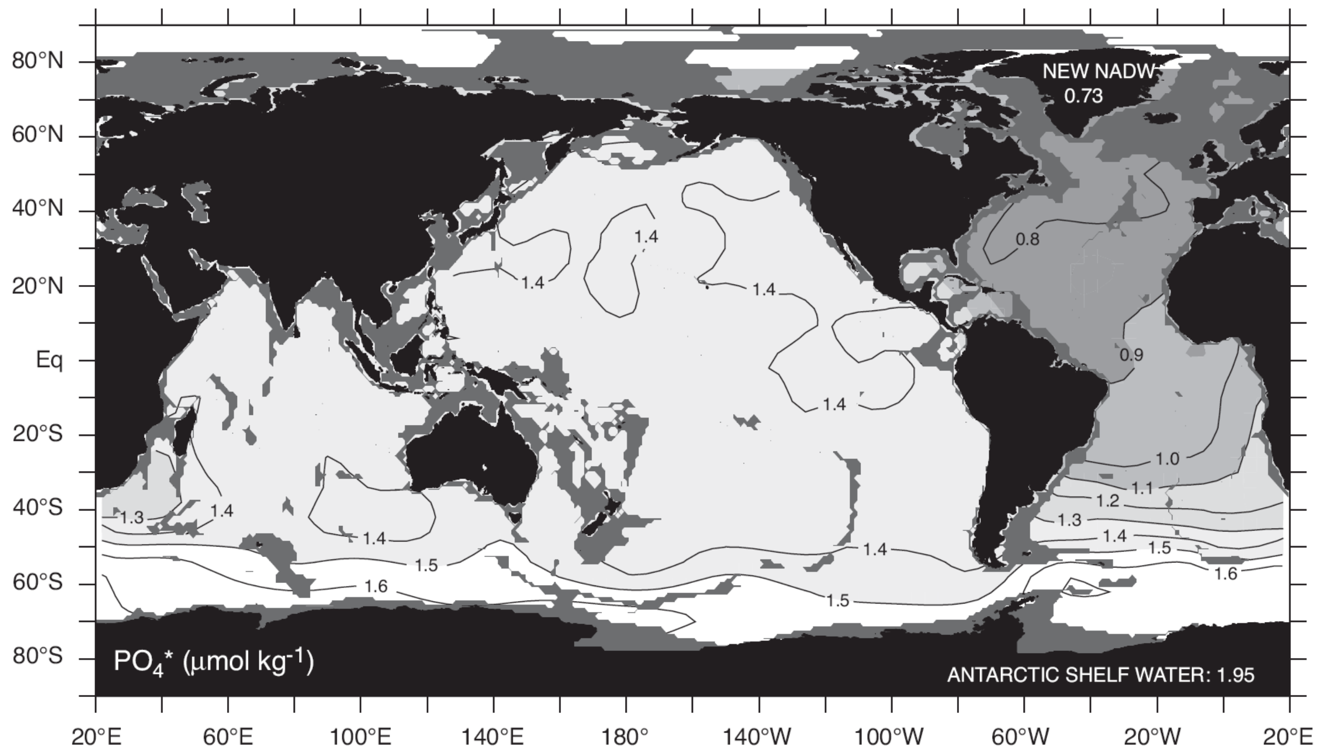


Fig. 4. Map showing percentage of northern component water along the 3200 m depth horizon. The light lines define the geographic boundaries of this horizon.

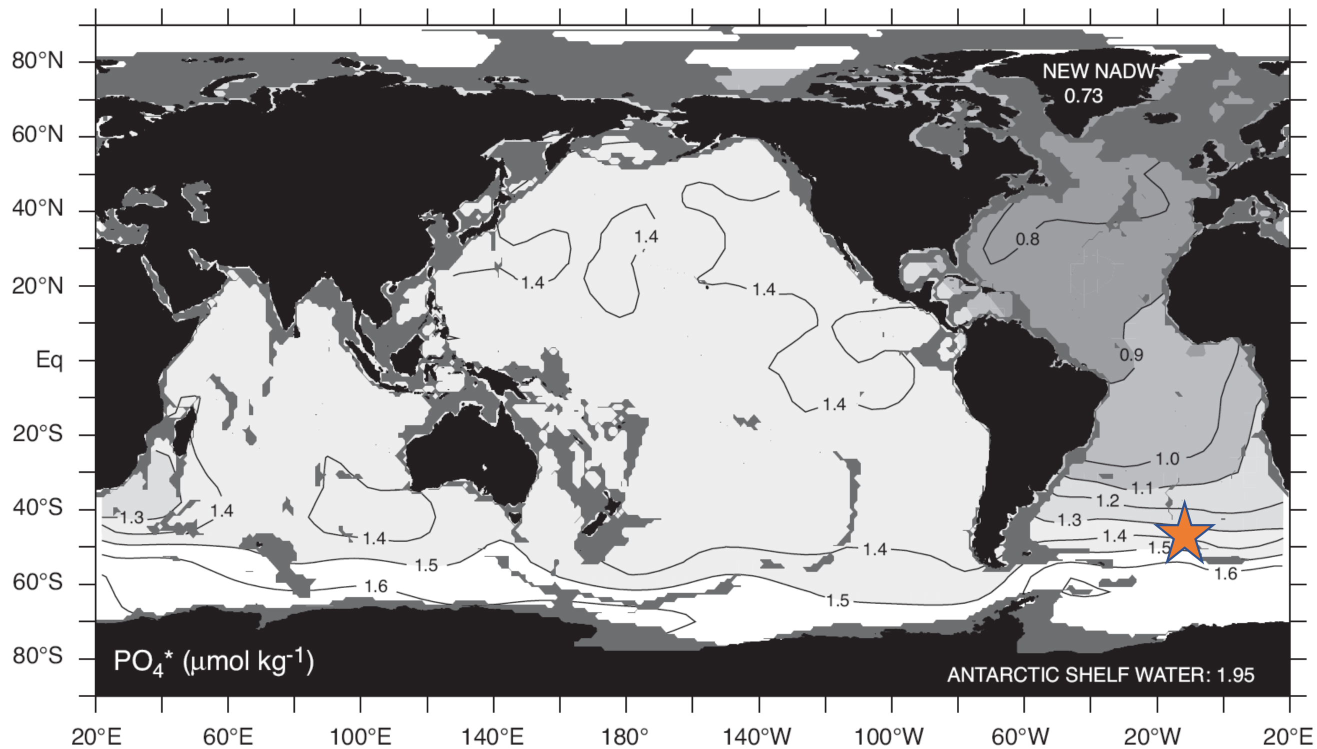
Broecker et al., 1991

PO_4^* distribution at 3000 m in the world ocean.



Sarmiento and Gruber, 2006

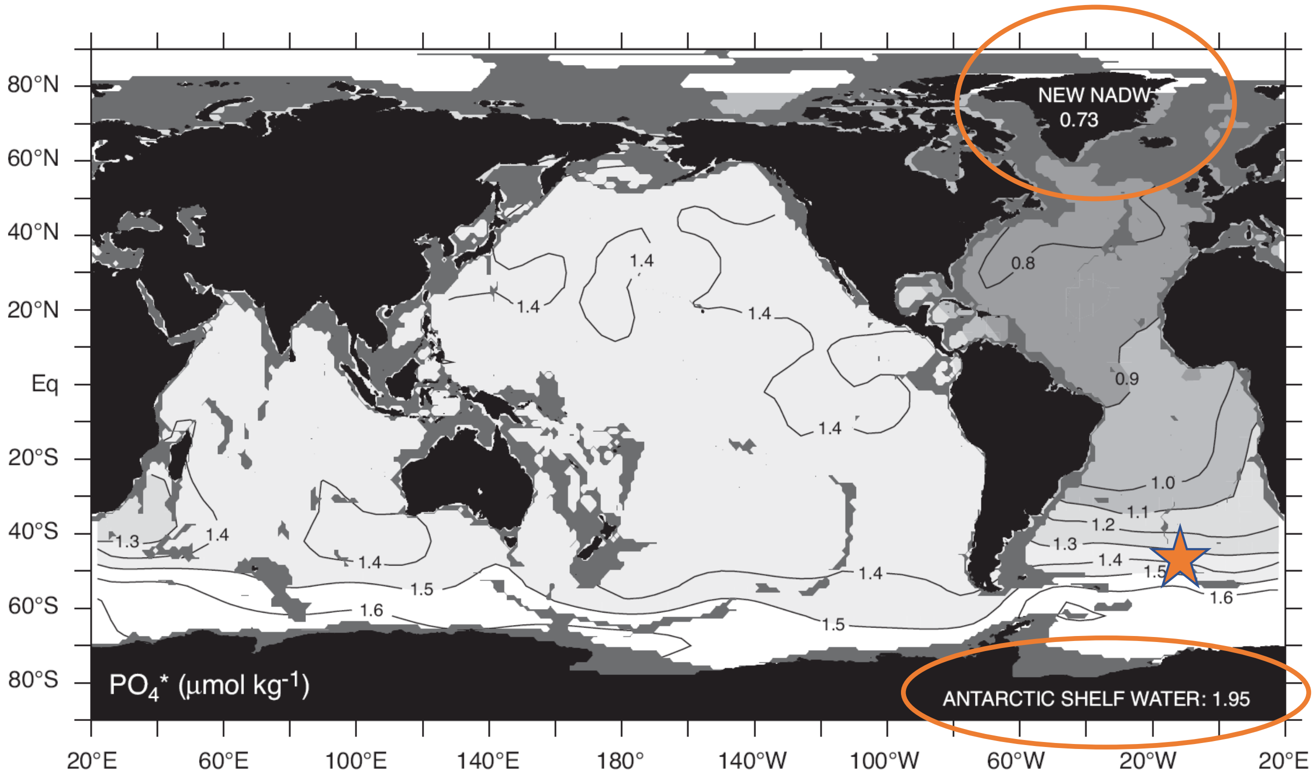
PO_4^* distribution at 3000 m in the world ocean.



PO4* at Orange star = 1.4. What are the relative fractions of NADW and Antarctic Shelf Water?

Sarmiento and Gruber, 2006

PO_4^* distribution at 3000 m in the world ocean.



$$f_{NADW} = \frac{1.95 - [PO_4^*]}{1.95 - 0.73} = \frac{1.95 - 1.4}{1.95 - 0.73} = 45\%$$

Sarmiento and Gruber, 2006

Quasi-conservative tracers depend on stoichiometry

Table 1.6. *Stoichiometric "Redfield" ratios for consumption of P, N, C and production of O_2 during photosynthesis and the opposite reaction during respiration in the ocean*

All values are relative to a phosphorus value of 1.0.

Source	Organic matter			O_2
	P	N	C	
Redfield *et al.*, 1963[a]	1.0	16	106	138
Anderson and Sarmiento, 1994[b]	1.0	16 ± 1	117 ± 14	170 ± 10
Anderson, 1995[c]	1.0	16	106	141–161
Kortzinger *et al.*, 2001[d]	1.0	17.5 ± 2.0	123 ± 10	165 ± 15
Hedges *et al.*, 2002[e]	1.0	17	106	154

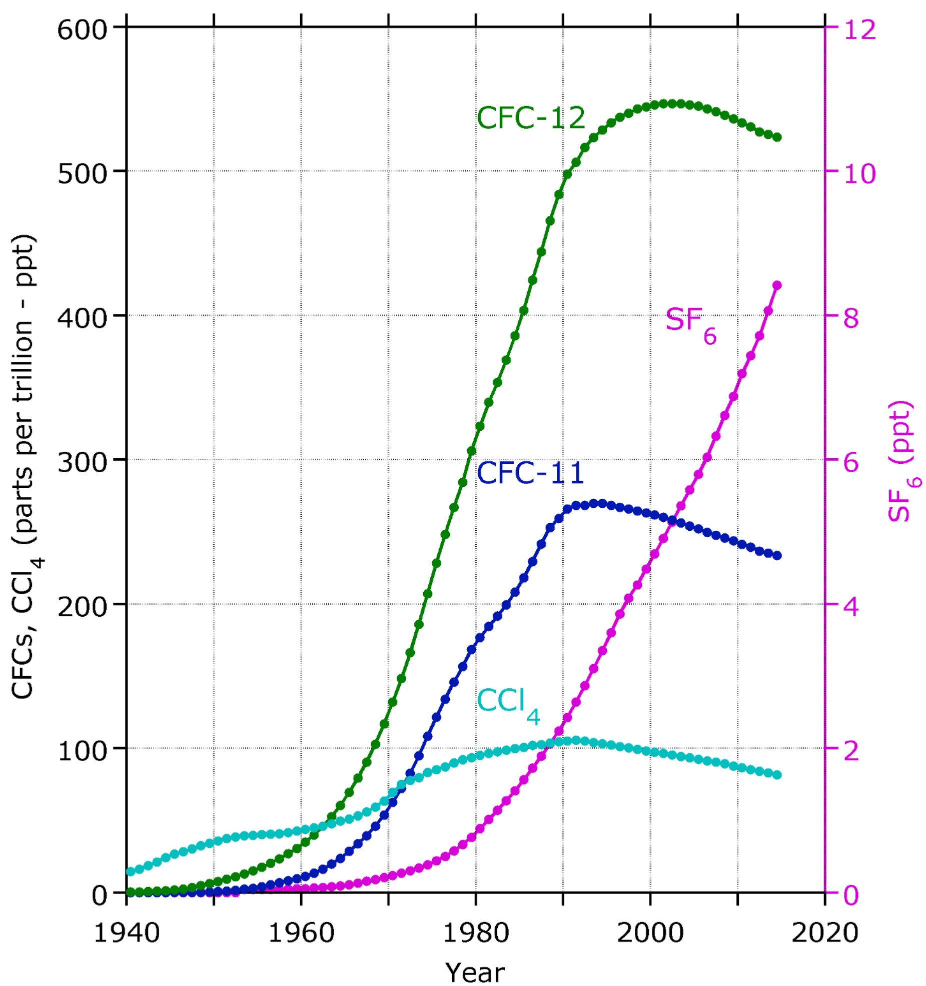
$\Delta O_2/\Delta N$	$\Delta O_2/\Delta P$
8.6	138
10.6	170
8.8	141
9.4	165
9.1	154

Emerson and Hedges, 2008

Transient Tracers

- Usually anthropogenic compounds with time-varying sources (or sinks)
  - CFCs, SF_6, ^{14}C, ^{3}H
- They enter the surface waters of the ocean either via gas exchange (e.g., CFCs, SF_6, ^{14}C) or water vapor exchange (^{3}H)
- Transient tracers allow:
  - Penetration of surface perturbations into the interior of the oceans to be visualized.
  - Delineation of pathways that newly formed subsurface water follows after leaving the surface.
  - Investigation of ocean circulation and mixing, and validation/calibration of GCMs.
  - Determination of water mass formation rates
  - Estimation of anthropogenic CO_2 inventory in the ocean

Northern Hemisphere Atmospheric Concentrations:
CFCs, CCl_4 and SF_6



- CFCs – Cholorfluorocarbons (refrigerant)
- CCl_4 – Carbon Tetracholoride (fumigant)
- SF_6 – Sulphur hexafluoride (semi-conductors)